

(Chapter 10 → Batch Processing)

Analogy of wage pipes.

(process 1 → | process 2 → | process 3)

Way in which ~~unix~~ UNIX operates is in ways like

store output of process 1 in stdout and stdin ready in for process 2

Basically output of 1 process is stored somewhere and transferred to input of another system.

But in UNIX we have only ~~Stream~~ 1 machine whereas MapReduce works by distributing it across multiple machines.

In UNIX $\text{stdin} \rightarrow \text{input}$ and $\text{stdout} \rightarrow \text{output}$
In Map Reduce read and write files
on distributed filesystem.

(HDFS, GFS) $\rightarrow$ mapreduce is built on
top of these file systems.

(Sort the URLs by their frequency count.)

Mapper $\rightarrow$ ~~reads~~ log file extracts the
url,
(url, 1). Then sends it to the
required ~~map~~ reducer

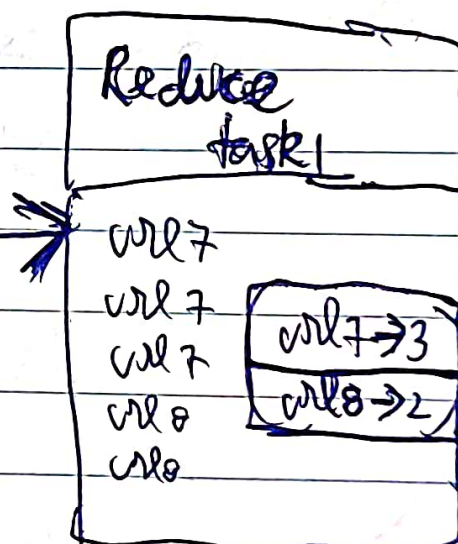
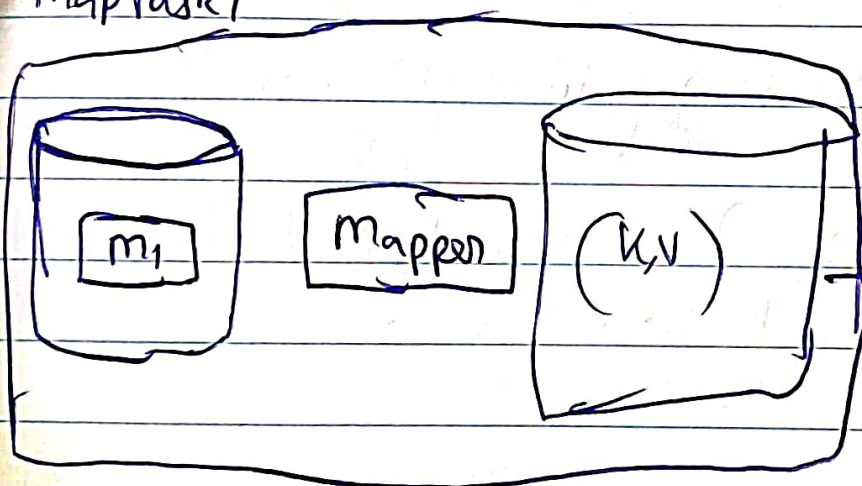
Reducer $\rightarrow$ aggregates these urls and completes
the step.

$\rightarrow$ 1st Map Reduce does the count part
Next map reduce to sort the urls by
count

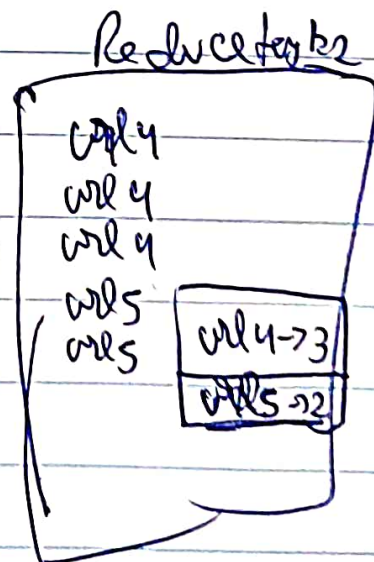
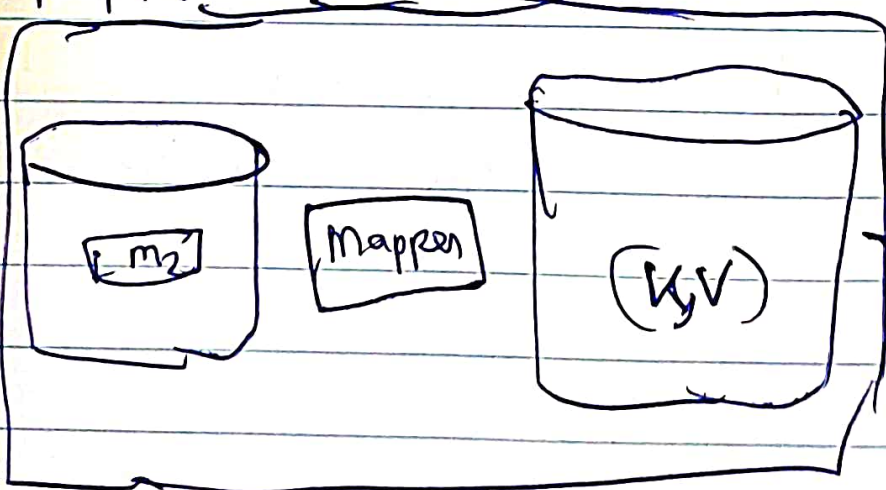
HDFS is smart it assigns the mappertasks to machines where the files are already residing.

(Increased Data locality, Putting computation near data, No Network call)

Map task 1



Map task 2



In UNIX the '|' pipe operator to pass output of one process to input of another using a small in-memory buffer.

Batch Processing

Map Reducers works on huge volumes of data, dividing work into small chunks (via mappers) and aggregated via (reducers)

Some use cases →

Build Search Indexes → After Map Reduce google can use and provide url to build search index

DataStore → we saw huge stream supplied to datastore for counting top k urls.

Avoid writing to DB → output is written to local disk of reducer, local network fast

(Few Limitations of Map Reduce)

- * Map Reduce saves immediate state in File (which allows fault tolerance but has ~~been~~ overhead to writing on disk)
- * Spark, Flink, defeat Map Reduce.
No immediate materialization,
For a single task multiple map reduce jobs might be required which now is being done by single spark job.